Pencil mathematical study: native preparation evidence

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Public synthetic known-answer mathematical research. Evidence observations only; no judgments, new cohort executions or qualification.

Zero executions refers to this named plan-only input, not historical experiments. Per-source software scopes are distinct and counts are never pooled.

Plan-only observations

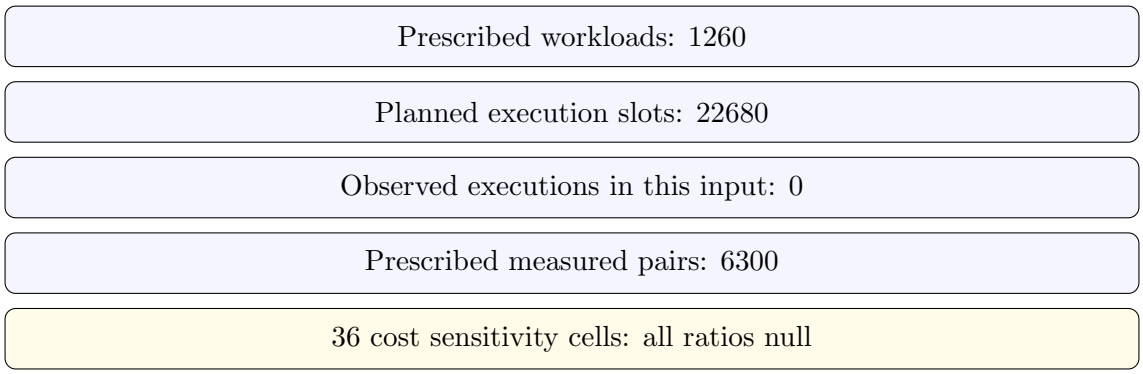


Diagram rather than a performance graph: this input contains no observed empirical ratios. Counts are read from source [plan]; no rates or uncertainty estimates are inferred. Warmup execution slots prescribed: 3780. No empirical cost ratio is available in this snapshot. Software validation and prepared inputs do not constitute independent protocol, binary, resource or launch admission.

Scoped source observations

[**first-gate**] Explicit user approval of larger FIRST gate only; cumulative budget, original deadline, IAM/controller, freeze and launch remain pending.
coefficient_cap: 25000000
seconds_per_execution: 3600
targets: 60
[**native-preparation**] Current native preparation software and run-free parity only; actual frozen archive reextracted with current source543 instrument. Does not independently reproduce the instrument build.
archive_reextracted: true
archive_workloads: 1260
cohort_slots: 22680
exact_retained_specs: true
software_tests: 3
solver_executions: 0
tests_failed: 0
[**native-canonical**] Native canonical packaging/protocol software fixtures only; distinct from empirical targets and previous producer/analyzer suites.
clippy_passed: true
empirical_targets: 0
qualified_cost_ratio: null
tests_failed: 0

tests_passed: 28
[**native-report**] Native renderer software fixtures only; includes pdflatex PDF test and historical validation-scope preservation.
clippy_exit_code: 0
empirical_targets: 0
tests_failed: 0
tests_passed: 8
[**instrument543**] Previously verified static ecbench build source543, separate from native tooling; plan-only checks and no solver execution.
archive_targets: 1260
prepared_plans: 9
solver_executions: 0
source: "543a54b4b1c3be7052872af3ffe393a44c69a3c3"
[**root-full**] Source0cd test-only correction validation lane: full default-parallel release suite failed. Exact source-file hashes are retained in receipt; focused passes do not replace this result.
exit_code: 101
failed: 2
ignored: 123
passed: 5275
[**root-focused**] Separate bounded serial reruns of the two failed tests from the exact same compiled binary; not a passing full-suite result.
allocator_failed: 0
allocator_passed: 1
exit_code: 0
gpu_failed: 0
gpu_passed: 1
[**canonical-hash**] Separate supplemental historical hash test, not a rerun or total of the 28-test suite. Earlier category-confusion failure retained; raw-file digest differs from canonical body digest by definition.
empirical_executions: 0
independent_bytes_match: true
runtime_source_changed: false
targeted_test_passed: 1

Evidence and limits

No software test count is an empirical target count. Missing costs stay null. Resource authorization is limited to its explicit source scope; no budget, deadline, infrastructure approval or independent review is inferred. The prior 19:06 report and its Python generator remain historical artifacts; new reports use this native renderer. [**plan**] named plan-only input; not historical experiments
`/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/validation/accounting/aggregate/actual-plan-only-summary-final.json`
SHA256: `9c0d055e2af7f6f79ce201b9b0c2a22691da0af55e8f6f127acec7070bb7e665`
[**first-gate**] Explicit user approval of larger FIRST gate only; cumulative budget, original deadline, IAM/controller, freeze and launch remain pending.
`/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/protocol_draft/accounting_amendment_20261029/first_gate_authorization.json`
SHA256: `e04f26d0cab6850f3c477e9521ea4359c46f64ec436ce899b3b4049867710600`
[**native-preparation**] Current native preparation software and run-free parity only; actual frozen archive reextracted with current source543 instrument. Does not independently reproduce the instrument build.
`/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/native_preparation/validation/receipt.json`

SHA256: 46843c028008698dc4eaa47370191904cf274791402d8953e23868e725d62ea3
[**native-canonical**] Native canonical packaging/protocol software fixtures only; distinct from empirical targets and previous producer/analyzer suites.
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/native_canonical_export/validation/receipt.json
SHA256: aeef1f2bc1984f248121a786996cfb6ed5008750a825d47712e53092ad4ea0f6
[**native-report**] Native renderer software fixtures only; includes pdflatex PDF test and historical validation-scope preservation.
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/native_review_report/validation-layout.json
SHA256: 37da6488934ed19fb48102e16de777a0c7935519f36245bd31399e4708864023
[**instrument543**] Previously verified static ecbench build source543, separate from native tooling; plan-only checks and no solver execution.
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/validation/instrument_build_40938439022/receipt.json
SHA256: 59fd9e0e1f4573d9492bcca710b5351cd0175f788b197a7c82939e177ab1a1a1
[**root-full**] Source0cd test-only correction validation lane: full default-parallel release suite failed. Exact source-file hashes are retained in receipt; focused passes do not replace this result.
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/validation/hash_one_ci_20261029/correction.json
SHA256: 2c83885fbefef25323bf968aefdb1ea96c1b67bdd4ed7dcae84cb7c5d68014fe
[**root-focused**] Separate bounded serial reruns of the two failed tests from the exact same compiled binary; not a passing full-suite result.
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/validation/hash_one_ci_20261029/correction.json
SHA256: 2c83885fbefef25323bf968aefdb1ea96c1b67bdd4ed7dcae84cb7c5d68014fe
[**canonical-hash**] Separate supplemental historical hash test, not a rerun or total of the 28-test suite. Earlier category-confusion failure retained; raw-file digest differs from canonical body digest by definition.
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/native_canonical_export/validation/historical-hash-receipt.json
SHA256: 6fe3e515c1c2a5ba81116d4c93ea95e16313fc2781e9fbf19c8c0180a94dc6c2